

Let it Calm: Exploratory Annealed Decoding for Verifiable Reinforcement Learning

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Abstract

Reinforcement learning with verifiable rewards (RLVR) is a powerful paradigm for enhancing the reasoning capabilities of large language models (LLMs), yet its success hinges on effective exploration. An ideal exploration strategy must navigate two fundamental challenges: it must preserve sample quality while also ensuring training stability. While standard fixed-temperature sampling is simple, it struggles to balance these competing demands, as high temperatures degrade sample quality and low temperatures limit discovery. In this work, we propose a simpler and more effective strategy, Exploratory Annealed Decoding (EAD), grounded in the insight that exploration is most impactful on early tokens which define a sequence’s semantic direction. EAD implements an intuitive *explore at the beginning, exploit at the end* strategy by annealing the sampling temperature from high to low during generation. This dynamic schedule encourages meaningful, high-level diversity at the start, then gradually lowers the temperature to preserve sample quality and keep the sampling distribution close to the target policy, which is essential for stable training. We demonstrate that EAD is a lightweight, plug-and-play method that significantly improves sample efficiency, consistently outperforming fixed-temperature sampling across various RLVR algorithms and model sizes. Our work suggests that aligning exploration with the natural dynamics of sequential generation offers a robust path to improving LLM reasoning.

1 Introduction

Reinforcement learning with verifiable rewards (RLVR) is a powerful approach to enhance the capabilities of Large Language Models (LLMs), particularly in mathematical reasoning (OpenAI, 2024; Guo et al., 2025; Team et al., 2025; Yang et al., 2025). In this framework, an LLM learns by iteratively generating potential solutions (i.e., rollouts), and receiving feedback on its attempts. A central challenge lies in guiding language models to explore diverse yet high-quality solutions in their vast output space (Cheng et al., 2025); this reflects the long-standing hard trade-off between exploration and exploitation in RL (Thrun, 1992; Sutton et al., 1998).

To achieve effective exploration, the sampling process can be modified to increase the variance of its underlying distribution. However, any such modification involves two fundamental challenges.

First, it must **preserve sample quality**. Increasing diversity at the cost of generating low-quality, nonsensical outputs is counterproductive. Second, it must **ensure training stability**. Modifying the sampler creates a discrepancy between the behavior policy (used for sampling) and the target policy (being optimized), necessitating an importance sampling (IS) correction in the gradient update (Degris et al., 2012). Large IS probability ratios can cause high-variance gradients that destabilizes the entire training process (Schulman et al., 2017). An ideal exploration technique must therefore increase diversity while keeping the sampling distribution close enough to the target policy to allow for stable learning (Haarnoja et al., 2018; Ziegler et al., 2019).

A principled approach to this trade-off is adjusting the sampling temperature (Ackley et al., 1985; Hou et al., 2025), which is both simple and *variationally optimal* (i.e., maximizing entropy and thereby increasing diversity while bounding KL divergence from the target policy (Jaynes, 1957)). However, relying on a single fixed temperature creates a tension: high temperature promotes diversity but produces nonsensical text (Renze, 2024; Wang et al., 2025b), whereas low temperature improves quality but limits exploration, yielding generic and repetitive outputs (Holtzman et al., 2020; Guo et al., 2025).

In this work, we propose **Exploratory Annealed Decoding (EAD)**, a strategy that improves the balance of this trade-off by leveraging a key insight into sequential generation: exploration is not equally valuable at every step. The initial tokens shape a sequence’s semantic direction and structure, making early exploration crucial for discovering diverse valid solutions. Later tokens, however, fill in details within the established context, where excessive exploration can harm coherence. This insight motivates our core strategy: *explore at the beginning, exploit at the end*. This simple principle elegantly addresses the twin challenges of quality and stability. Injecting randomness early promotes diverse, high-level exploration, while reducing it later ensures completions are both coherent and close to the target policy—an essential property for stable off-policy learning.

In summary, our contributions are as follows:

- ① We propose EAD, a simple and effective exploration strategy for RLVR that dynamically anneals temperature to encourage meaningful diversity while maintaining high sample quality.
- ② We show EAD is a plug-and-play enhancement that improves sample efficiency over temperature sampling, delivering robust gains across various RLVR algorithms including GRPO (Shao et al., 2024), DAPO (Yu et al., 2025), and EntropyMech (Cui et al., 2025) on both small and larger models.
- ③ We show that EAD can be adapted for test-time inference, where a tuned temperature schedule further enhances generation quality.

2 Preliminary

Notations. Let x be a prompt from a dataset $\mathcal{D}$, and let $y = (y_1, \dots, y_{|y|})$ be a generated response sequence, where $y_{<t}$ denotes the prefix $(y_1, \dots, y_{t-1})$. We define an LLM as a policy π_θ parameterized by θ , and denote the reference policy, i.e., the starting point for RL, as π_{ref} . The probability of generating y given x is defined autoregressively as $\pi_\theta(y \mid x) = \prod_{t=1}^{|y|} \pi_\theta(y_t \mid [x, y_{<t}])$. A reward model $R(x, y)$ evaluates the quality of a prompt-response pair; for our RLVR experiments, we use the rule-based Math-Verify reward model.¹ Finally, we use $|\cdot|$ to denote the length of a sequence or the cardinality of a set, and use the shorthand $1 : n$ for the set $\{1, \dots, n\}$.

Reinforcement Learning with Verifiable Rewards (RLVR). The standard objective for Reinforcement fine-tuning of LLMs is to maximize the expected reward over a prompt dataset $\mathcal{D}$:

$$\max_{\theta} J(\theta) := \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(\cdot \mid x)} [R(x, y)]. \quad (1)$$

However, on complex reasoning tasks, learned reward models $R(x, y)$ are prone to *reward hacking*, where they assign high scores to plausible but incorrect solutions (Gao et al., 2023; Perez et al., 2023; Weng, 2024; Wang et al., 2025a). RLVR addresses this by replacing the learned model with a verifiable, rule-based reward signal—such as a verifier that provides binary feedback on a solution’s correctness (Guo et al., 2025). This ensures that the policy is optimized using a reliable signal.

RLVR is commonly implemented using policy gradient algorithms like Proximal Policy Optimization (PPO) (Schulman et al., 2017). However, standard PPO often requires complex token-level advantage estimation and a separate value model. To better suit RLVR’s trajectory-level binary rewards, subsequent methods simplify the advantage calculation

¹<https://github.com/huggingface/Math-Verify>

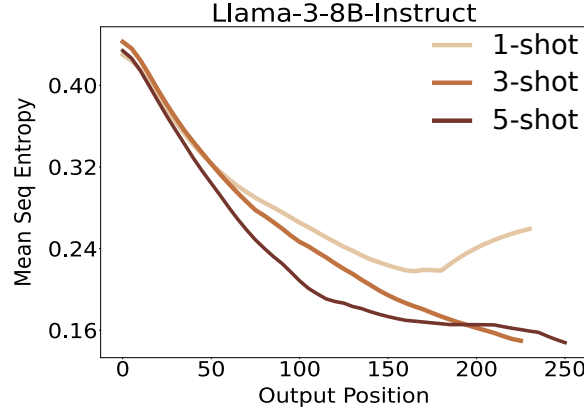


Figure 1: Average entropy shrinks with output positions for Llama-3-8B-Instruct on MMLU dataset.

(Shao et al., 2024; Yu et al., 2025). A prominent example is Decoupled Clip and Dynamic Sampling Policy Optimization (DAPO) (Yu et al., 2025), which optimizes:

$$J_{\text{DAPO}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, y^{(1:G)} \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{\sum_{i=1}^G |y^{(i)}|} \times \sum_{i=1}^G \sum_{t=1}^{|y^{(i)}|} \min \left\{ r_t^{(i)}(\theta) A_i, \text{clip} \left(r_t^{(i)}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}} \right) A_i \right\} \right]$$

s.t. $0 < |\{y^{(i)} : y^{(i)} \text{ is correct}\}| < G,$

where $\pi_{\theta_{\text{old}}}$ refers to previous policy and $r_t^{(i)}(\theta) = \frac{\pi_{\theta}(y_t^{(i)} | [x, y_{<t}^{(i)}])}{\pi_{\theta_{\text{old}}}(y_t^{(i)} | [x, y_{<t}^{(i)}])}$. The asymmetric bound $\varepsilon_{\text{high}} > \varepsilon_{\text{low}}$ is proposed to relax the restriction on probability increase and encourage exploration in the training. The advantage A_i is computed by normalizing the binary rewards across a batch of G responses, thus avoiding the need for a value model:

$$A_i = \frac{R_i - \text{mean}_{k \in 1:G}(R_k)}{\text{std}_{k \in 1:G}(R_k)}, \text{ where } R_i = R(x, y^{(i)}).$$

Temperature. Temperature sampling (Ackley et al., 1985) is a widely used method to control the stochasticity of the policy π_{θ} . At each generation step t , the LLM computes a vector of logits, $\mathbf{h}$, over the vocabulary V based on the prompt x and the preceding tokens $y_{<t}$. Temperature sampling rescales these logits with a parameter $\tau > 0$ before applying the softmax function to form the next-token probability distribution:

$$\pi_{\theta}(y_t = v | [x, y_{<t}]; \tau) = \frac{\exp(h_v / \tau)}{\sum_{v' \in V} \exp(h_{v'} / \tau)},$$

where v is a token in the vocabulary V and h_v is its corresponding logit. The temperature τ directly modulates the sharpness of the output distribution. A higher temperature ($\tau > 1$) flattens the distribution, increasing output diversity by making less likely tokens more probable. Conversely, a lower temperature ($\tau < 1$) sharpens it, leading to more deterministic, greedy outputs.

Pass@k. Pass@ k measures the probability that at least one of k independent outputs from a language model is correct. Let p_x denote the underlying accuracy of the language model π given one prompt x . The pass@ k accuracy is defined as $\mathbb{E}_{x \sim \mathcal{D}} [1 - (1 - p_x)^k]$. The inner term $1 - (1 - p_x)^k$ quickly approaches one unless p_x is near zero. For example, when $k = 64$, any $p_x \geq 0.0695$ already yields a probability of at least 0.99. A large gap between pass@1 and pass@ k suggests that for some prompts, p_x is essentially zero. High diversity may benefit this metric because a model with diverse outputs is more likely to maintain non-negligible p_x values across prompts, leading to stronger pass@ k performance.

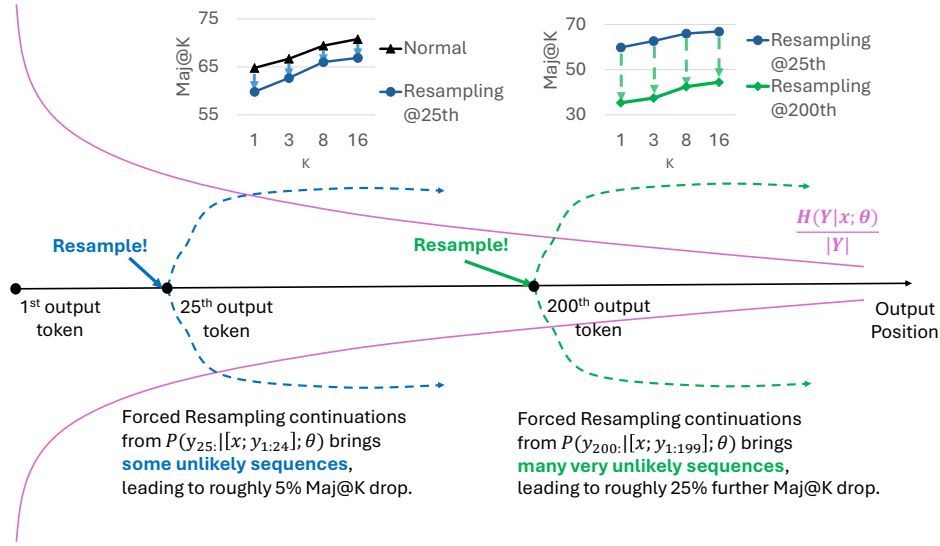


Figure 2: A “forking” experiment on DeepSeek-Llama-3 shows early branching (high-entropy region) yields higher Maj@k on MMLU than late branching (low-entropy region).

113 **Entropy.** Given a prompt x and a prefix $y_{<t}$, the **token-level entropy** at step t is defined
 114 over the policy’s conditional distribution:

$$H(Y_t \mid [x, y_{<t}]; \theta) := - \sum_{v \in V} \pi_\theta(y_t = v \mid [x, y_{<t}]) \log \pi_\theta(y_t = v \mid [x, y_{<t}]),$$

115 where the sum is over all tokens v in the vocabulary V .

The **average entropy** of the policy, $H(\pi_\theta)$, is the expected token-level entropy over all prompts and generation steps. We can estimate this value empirically using Monte Carlo sampling. For each prompt $x \in \mathcal{D}$, we generate G i.i.d. responses $y^{(1)}, \dots, y^{(G)}$. The average entropy is then approximated by averaging the token-level entropies across all generated tokens:

$$\bar{H}(\pi_\theta) \approx \frac{1}{|\mathcal{D}|} \sum_{x \in \mathcal{D}} \left(\frac{\sum_{i=1}^G \sum_{t=1}^{|y^{(i)}|} H(Y_t \mid [x, y_{<t}^{(i)}]; \theta)}{\sum_{i=1}^G |y^{(i)}|} \right).$$

116 3 Sequential Exploration: Explore Early, Exploit Late

117 Exploration is a cornerstone of reinforcement learning, enabling agents to discover high-
 118 quality policies rather than settling on suboptimal solutions (Sutton et al., 1998; Ladosz
 119 et al., 2022). This principle becomes particularly vital in deep RL, where vast action spaces
 120 render exhaustive search infeasible. In the context of RL for language models (e.g., RLVR),
 121 insufficient exploration often manifests as *entropy collapse*, i.e., a premature narrowing of
 122 the generation distribution during training (Yu et al., 2025; Wang et al., 2025b; Cui et al.,
 123 2025). A common simple tool to encourage exploration is *temperature sampling*. However,
 124 a *fixed* temperature imposes a difficult trade-off. A high temperature promotes diversity
 125 (as indicated by increased entropy²), but it risks degrading output quality with nonsensical
 126 tokens and hallucinations (Renze, 2024; Wang et al., 2025b). In contrast, a low temperature
 127 limits the discovery of novel solutions, leading to generic and repetitive outputs (Holtzman
 128 et al., 2020; Guo et al., 2025). The key to resolving this dilemma lies not in finding a
 129 single best temperature, but in recognizing that exploration requirements vary throughout
 130 the generation process. This key insight stems directly from the autoregressive nature of
 131 language models. At the beginning of a sequence, the context is minimal and uncertainty

²See § C for the proof.

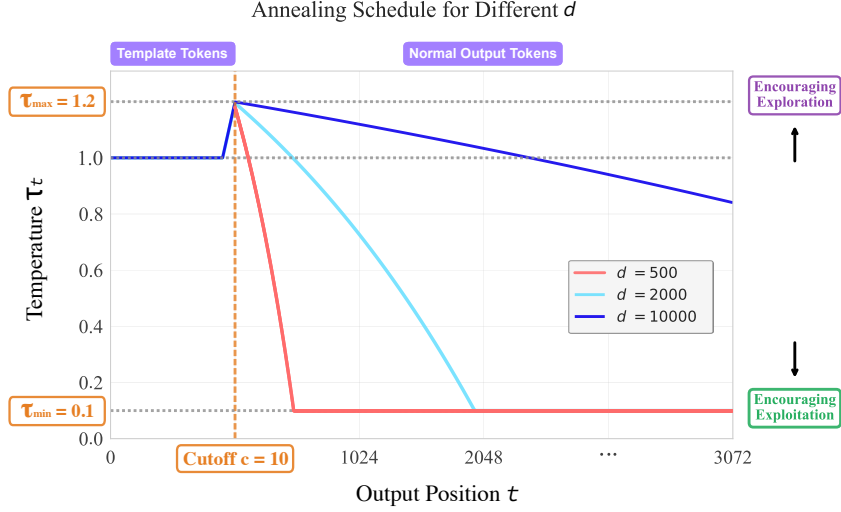


Figure 3: The annealing schedule with different decay rates d . A larger d slows the cooling, front-loading exploration over more tokens. We set $c = 10$, $\tau_{\max} = 1.2$, $\tau_{\min} = 0.1$ for illustration.

is high, allowing a wide range of valid continuations. As more tokens are produced, the context becomes increasingly specific, constraining subsequent choices. We reference the Data Processing Inequality (DPI) (Shannon, 1948) as a *theoretical motivation* to investigate these entropy dynamics. While not a strict derivation for language models, the DPI provides a framework to hypothesize that expected conditional entropy tends to decrease as the context grows:³

$$\begin{aligned}
 H(Y_t | [x, Y_{<t}]; \theta) &= \underbrace{\mathbb{E}_{y_{<t}} [H(Y_t | [x, y_{<t}]; \theta)]}_{\substack{\text{Expected entropy at step } t \\ \text{(average over all prefixes } y_{<t})}} \\
 &\geq \underbrace{\mathbb{E}_{y_{<t+1}} [H(Y_{t+1} | [x, y_{<t+1}]; \theta)]}_{\substack{\text{Expected entropy at step } t+1 \\ \text{(average over all prefixes } y_{<t+1})}} \\
 &= H(Y_{t+1} | [x, Y_{<t+1}]; \theta)
 \end{aligned}$$

Fig. 1 presents an empirical validation, examining position-wise entropy trend on MMLU (Hendrycks et al., 2021)⁴ with Llama-3-8B-Instruct (Grattafiori et al., 2024).

This entropy decay has a direct performance implication: effective exploration, producing diverse, high-quality responses, is most beneficial in an uncertain, high-entropy phase at the start of generation. To verify this, we replicate a controlled “forking” experiment (Yang & Holtzman, 2025) on DeepSeek-Distilled Llama-3-8B (Guo et al., 2025), a representative RLVR-fine-tuned model from the same Llama-3 family. As shown in Fig. 2, trajectories branched from early generation steps consistently outperform those branched later. This finding is also consistent with observations from inference-time analysis, where forced, late-stage exploration tends to degrade output quality (Liao et al., 2025; Yang & Holtzman, 2025; Fu et al., 2025).

Aligning our strategy with the natural dynamics of generation, we arrive at a simple yet powerful design principle: **explore early and exploit late**. We emphasize that this is a practical heuristic grounded in the empirical entropy dynamics of autoregressive models, not a claim that late-stage exploration is never useful. Individual sequences may contain critical late-stage decisions; however, in expectation, the marginal value of exploration

³While specific rollouts may have late high-entropy positions, the probability of this is exponentially small with position t (Yang & Holtzman, 2025), making the overall trend a reliable heuristic.

⁴We use MMLU as a held-out dataset with Chain-of-Thought prompting (Wei et al., 2022) to incentivize longer reasoning outputs, aligning with a typical RLVR scenario.

is highest at the beginning. Moreover, EAD does not eliminate late-stage stochasticity entirely—the floor temperature $\tau_{\min} > 0$ ensures that a non-trivial degree of randomness is retained throughout generation.

4 A Method for Sequential Exploration

To put the principle of “explore early, exploit late” into practice, we introduce *Exploratory Annealed Decoding (EAD)*, which uses an annealed temperature schedule starting from a higher-than-standard initial temperature (i.e., $\tau > 1$). To adapt this strategy to RLVR, we further incorporate a *global-step-aware decay rate*, ensuring that the temperature schedule remains effective as the typical response length increases during training.

Exploratory Annealed Decoding. Instead of a fixed temperature, our method dynamically adjusts the temperature τ_t for each token t in a rollout. The schedule starts at a high temperature $\tau_{\max} > 1$ and decreases progressively throughout the generation process. Specifically, we sample the t -th token for one rollout with the token-level temperature $\tau_t = \max\{1 + \tau_{\max} - e^{t/d}, \tau_{\min}\}$, where we apply the annealed schedule with a *decay rate* d controlling the annealing speed. As illustrated in Fig. 3, the decay rate d controls how long the policy remains in a high-exploration state. A larger d front-loads exploration across more initial tokens, while a smaller d transitions to exploitation more quickly. In practice, we let $\tau_t = 1.0$ for $t < c$, where c is a pre-determined initial position for the sake of model-specific or prompt-specific template tokens injected in the training process. During RLVR, language models tend to generate some template tokens such as “let’s verify step by steps” or repeat the question. We fix the temperature at $\tau = 1.0$ in this part to avoid interfering with the generation process. Typically, c is very small (e.g., a few dozen tokens corresponding only to the prompt or template length) and does not affect the core reasoning steps where exploration is most critical.

Global-Step-Aware Decay Rate. As training progresses and response lengths increase,⁵ the decay rate d should be adjusted in accordance with the training step. Otherwise, an excessive number of tokens may be generated under extremely low temperatures, which degrades response quality and leads to undesirable behaviors such as repetition (Guo et al., 2025), off-topic drift (Spataru et al., 2024), and unnecessary verbosity (Holtzman et al., 2020). In particular, we adopt the following *global-step-aware decay rate*: $d_s = \min(d_0 + \alpha \times s, d_{\max})$, where $\alpha > 0$ is the growth factor and $d_{\max}$ is the decay cap.

Ensuring Stability with Truncated Importance Sampling. With aggressive annealing schedules (e.g., very small $\tau_{\min}$ and d), sampling low-probability, long-tail tokens can cause the annealed policy to deviate significantly from the one being optimized. This creates an off-policy discrepancy that risks training instability. To mitigate this, we employ *truncated importance sampling* (TIS) (Heckman et al., 1998; Hilton et al., 2022; Yao et al., 2025) to correct the objective, ensuring stable optimization even under highly exploratory schedules (see § B for details).

Overall, this annealed decoding strategy offers a compelling combination of effectiveness and efficiency. As a plug-and-play modification to standard temperature sampling, it incurs negligible computational overhead and is fully compatible with existing RLVR pipelines and diverse policy optimization algorithms like DAPO, GRPO, etc.

5 Experiments

5.1 Experimental Setup

Models, Data, and Training Frameworks. To ensure a rigorous and controlled comparison, we follow the Minimal-RL recipe (Xiong et al., 2025),⁶ training all models on the Numina-Math dataset (Beeching et al., 2024), which contains 860k math prompts. To assess

⁵We illustrate increased length for EAD in § D.

⁶More training details and hyperparameter setups are illustrated in § A.

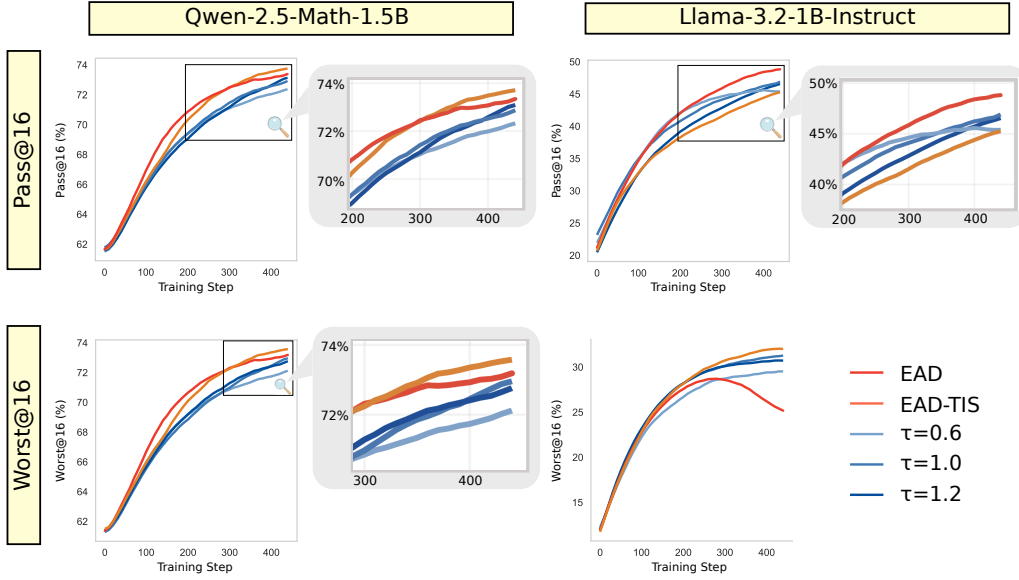


Figure 4: Pass@16 and Worst@16 performance evaluation in RL training. While EAD improves exploration of high-quality samples (even the worst outperform temperature sampling), the gain diminishes over time; importance sampling can supplement to correct bias and sustain training.

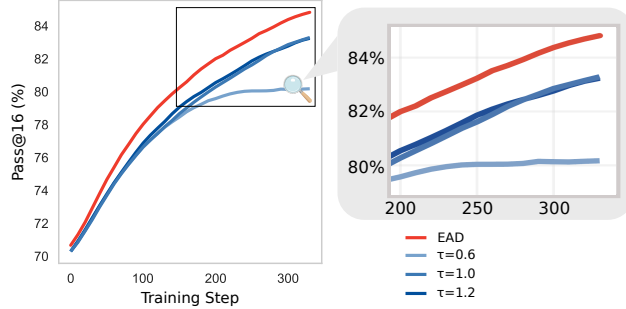


Figure 5: Pass@16 performance on Qwen-2.5-Math-7B. EAD enables better exploration than fixed-temperature sampling, yielding sustained gains in Pass@16 throughout training.

the generality of our method, we experiment with both Qwen-2.5-Math-1.5B (Yang et al., 2024) and Llama-3.2-1B-Instruct (Dubey et al., 2024).⁷ We also include the larger Qwen-2.5-Math-7B model to evaluate how our approach scales. While our primary experiments are conducted within the DAPO framework (Yu et al., 2025), we demonstrate broader applicability by additionally integrating EAD with GRPO (Shao et al., 2024) and EntropyMech (Cui et al., 2025).

Baselines and Controlled Comparison. We evaluate EAD against fixed-temperature sampling, with temperatures $\tau \in \{0.6, 1.0, 1.2\}$ as recommended by prior work (Renze, 2024; Guo et al., 2025; Hou et al., 2025). For a fair comparison focused specifically on the sampling strategy, we disable two orthogonal techniques for all methods: (1) dynamic data sampling (Yu et al., 2025), to maintain a consistent training set, and (2) rollout length penalties, to avoid confounding the reward signal with length-based biases.

Hyperparameters and Tuning. Unless otherwise stated, we use a default configuration of $\tau_{\max} = 1.2$, $d_0 = 25$, $\alpha = 5$, and $d_{\max} = 40000$ for EAD. For $\tau_{\min}$, we observed optimal values varied by model capability. For the 1B and 1.5B models, we set $\tau_{\min} = 0.1$. For the more capable 7B model, we used a higher value of $\tau_{\min} = 0.8$, as stronger models

⁷We also experimented with the Llama-3.2-1B base model. However, consistent with Wang et al. (2025c), we found that applying RL to base models without intermediate domain-specific fine-tuning yields limited gains across all methods. We defer a deeper investigation to future work.

already have sharper next-token distributions; a moderately high floor temperature prevents overconfident, near-greedy decoding that produces plausible but incorrect answers. The key benefit of EAD still holds: early tokens are sampled at a higher temperature than later ones, preserving the “explore early, exploit late” dynamic even within this narrower range. All hyperparameters, including the learning rates for both EAD and the fixed-temperature baselines, were carefully tuned based on a prior study over held-out datasets to ensure a fair comparison. We also provide a detailed ablation study on the sensitivity of these hyperparameters in § 5.5 to demonstrate the robustness of EAD.

5.2 EAD Improves RLVR Training

EAD Improves RL Exploration and Training Efficiency. As shown in Fig. 4, EAD significantly improves training efficiency. For Pass@16 accuracy, EAD (w/o TIS) consistently outperforms the baselines on the Llama and Qwen models (EAD (w/ TIS) also outperforms on the Llama model), demonstrating more effective exploration. Under the stricter Worst@16 metric, the inclusion of TIS becomes essential for maintaining stable performance gains, highlighting its importance for correcting the off-policy training dynamic introduced by EAD. Through bootstrapping evaluation as in Hochlehnert et al. (2025), the standard deviation of both Pass@16 performance and Worst@16 are way below 0.01 and thus all comparisons here are significant.

To verify that our method generalizes, we evaluated it on the larger Qwen-2.5-Math-7B model. The results, presented in Fig. 5, confirm that the performance gains from EAD remain significant. This demonstrates that our approach is effective not only on smaller models but also scales successfully.

EAD Mitigates Entropy Collapse. One major problem in RLVR training is entropy collapse (Cui et al., 2025), which shrinks the exploration space and constrains improvement during the “plateau stage” (Deng et al., 2025). We plot the entropy dynamic in Fig. 8, where we can see that the entropy dynamic for EAD-empowered methods is not monotonically decreasing from the beginning. Instead, it tries to gradually transition out from local optimum (Kirkpatrick et al., 1983; Bertsimas & Tsitsiklis, 1993) in a natural, continuous way without any external intervention, such as introducing tree search in rollout sampling (Li et al., 2025).

Sample efficiency of EAD. Increasing the number of rollouts is a common but computationally expensive strategy to enhance exploration (Hou et al., 2025). We test the sample efficiency of EAD by varying the number of rollouts, adjusting the learning rate accordingly as suggested by prior work (Chen et al., 2025). As shown in Fig. 9a, while more rollouts can further improve performance, EAD achieves strong results with just 4 or 8 rollouts. For instance, the optimal configurations are $n = 8$ with a learning rate of 10^{-6} for Llama-3.2-1B-Instruct and 2×10^{-6} for Qwen-2.5-Math-1.5B. This highlights the sample efficiency of our approach, offering a way to reduce the computational cost of the rollout phase.

Disentangling Length and Quality. A natural consequence of improved exploration is that EAD can induce slightly longer reasoning chains (see § D for a detailed comparison of response lengths). However, several lines of evidence strongly suggest that the gains are primarily driven by the intra-sequence temperature schedule rather than simply from generating longer outputs: (1) the reverse annealing schedule (Fig. 10), which also injects randomness and can produce longer responses, fails to match even standard temperature sampling; and (2) EAD improves inference-time Majority@N on off-the-shelf models (Fig. 7), where no RL-induced length increase is present.

To assess whether EAD’s advantage persists with extensive exploration, we compare it against baselines using a larger set of 16 rollouts. Fig. 9b shows that although the relative performance gain diminishes, EAD still outperforms fixed-temperature baselines by a clear margin.

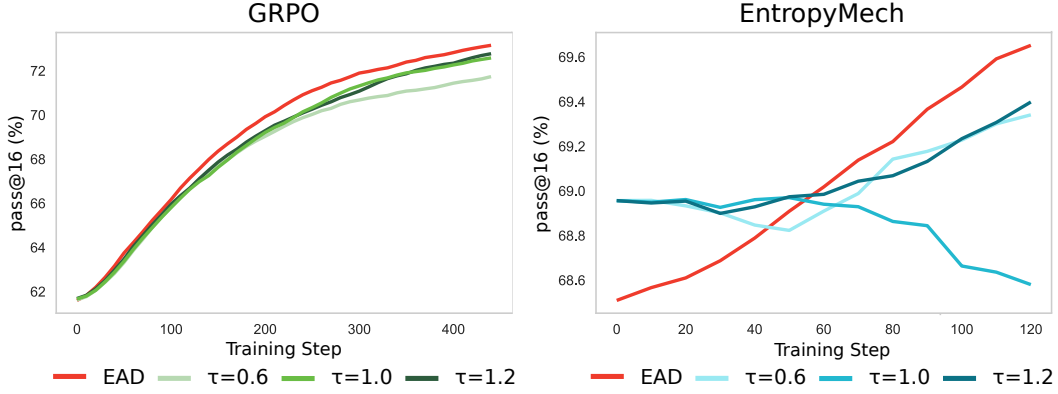


Figure 6: EAD is compatible with various RL algorithms and can significantly improve the model performance over time.

5.3 EAD is Compatible with Various RL Algorithms

To demonstrate that EAD is a general, plug-and-play exploration strategy, we evaluate its performance when integrated into two other prominent RL algorithms: GRPO (Shao et al., 2024) and EntropyMech (Cui et al., 2025). These algorithms provide diverse testbeds. GRPO is more conservative, constraining policy updates with a KL divergence penalty and stricter clipping mechanism that can limit exploration (Yu et al., 2025), while EntropyMech uses a specialized token-clipping mechanism to mitigate entropy collapse.

As shown in Fig. 6, EAD consistently outperforms fixed-temperature sampling in both frameworks. These results confirm the broad applicability of our method as an improved exploration strategy across different RL algorithms.

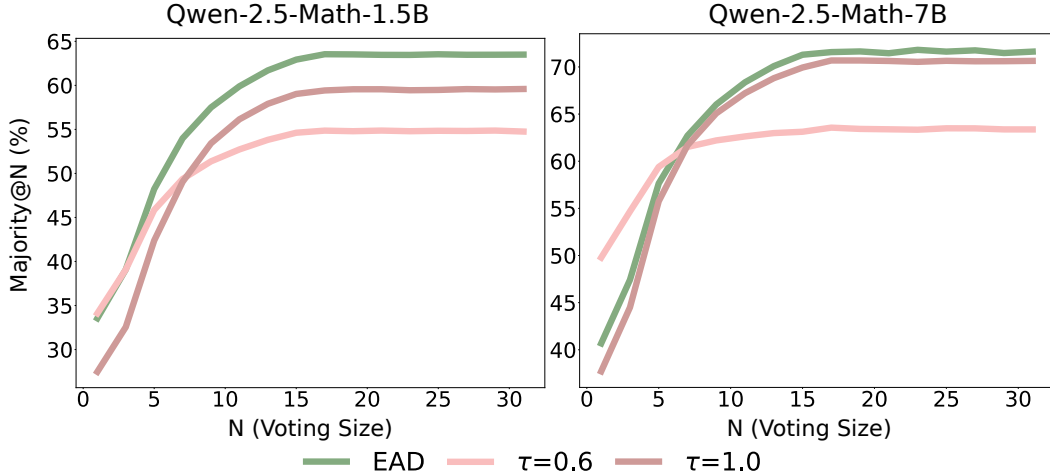


Figure 7: Inference-Time Scaling Evaluation for Different Decoding Methods using off-the-shelf Qwen2.5 models. We could see that EAD improves traditional temperature sampling. We set $\tau_{\max} = 1.2$, $\tau_{\min} = 0.1$, $d = 25$ for EAD.

5.4 EAD Improves Inference-Time Scaling

To understand whether the success of EAD in RL training is driven by its ability to generate high-quality samples, we conduct an evaluation at inference time. This experiment is designed to isolate the sampling strategy’s effectiveness from the dynamics of RL optimization (Berseeth, 2025). Using off-the-shelf Qwen-2.5 models without any fine-tuning, we compare EAD against fixed-temperature sampling. Since no oracle is available at inference time to select the single correct answer from N candidates, we adopt majority voting (Majority@N) (Wang et al., 2023; Snell et al., 2024), the standard practical aggregation

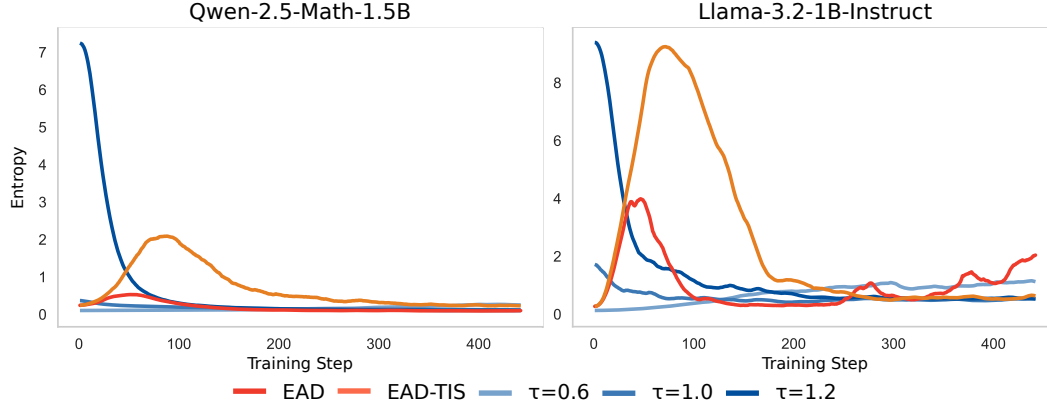
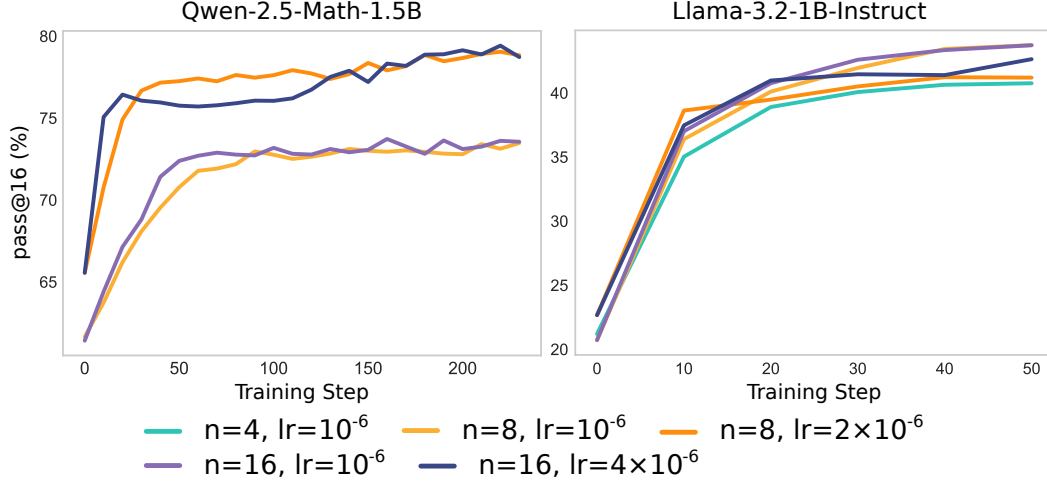
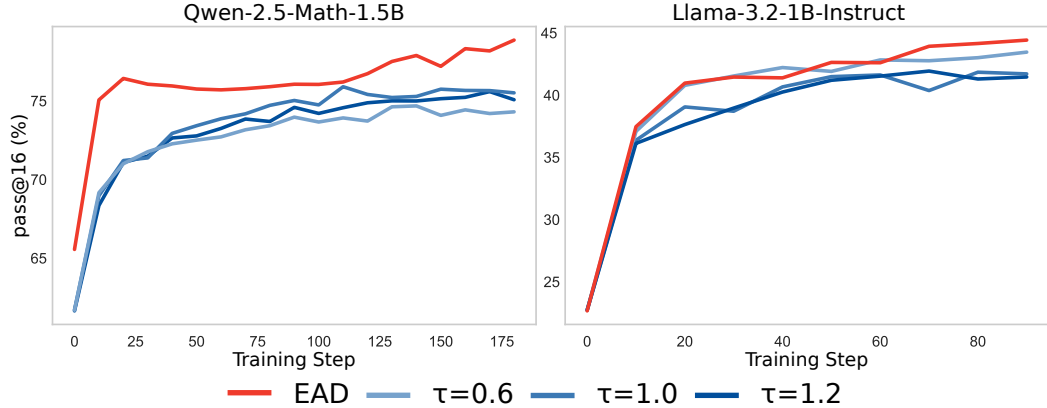


Figure 8: Entropy dynamics in RL training. With fixed temperature sampling, entropy decreases sharply from the start, limiting exploration. EAD helps the algorithm escape local minima by enabling exploration when needed during training.



(a) EAD would bring further performance improvement via increased numbers of rollouts, but the commonly used 4 or 8 is already good enough.



(b) When scaling out the rollout number to 16, the relative advantages of our methods diminished; however, it still outperforms traditional same-temperature sampling.

Figure 9: Sample Efficiency of EAD.

285 strategy for inference-time scaling. As shown in Fig. 7, EAD consistently improves over

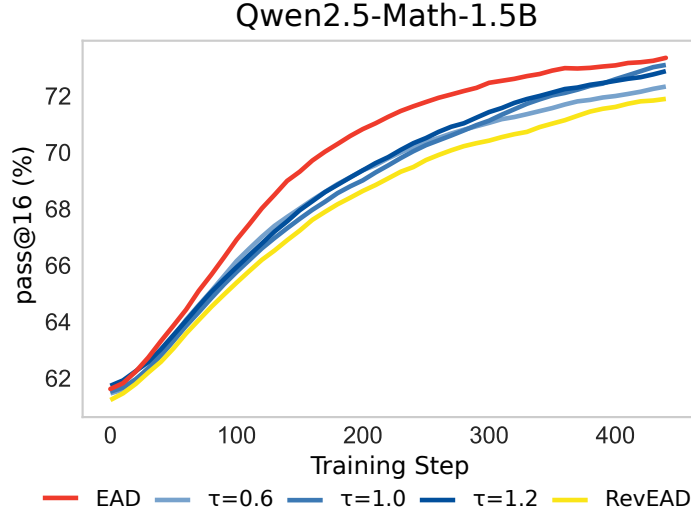


Figure 10: Reverse Annealing experiment results. The reverse schedule fails to surpass normal temperature sampling.

the baseline for most values of N . This result suggests that EAD’s temperature schedule produces samples that are not only diverse but also of sufficiently high quality to improve the majority-voted consensus, even without any RL training.

5.5 Ablation Study

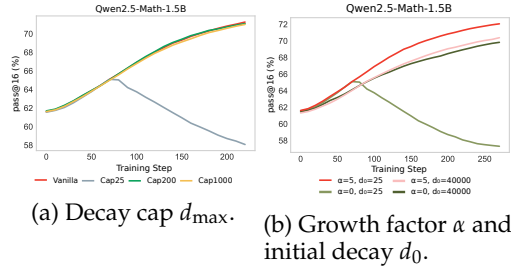


Figure 11: Ablation on hyperparameter sensitivity. **Left:** Ablation on decay cap $d_{\max}$. “Vanilla” denotes EAD with the default $d_{\max} = 40000$. Setting $d_{\max} = 25$ (where $d_0 = 25$) mimics a fixed temperature schedule (effectively $\alpha = 0$), leading to performance drops as the schedule fails to adapt to longer responses. EAD remains robust provided $d_{\max}$ allows for smooth temperature reduction (e.g., $d_{\max} \geq 200$). **Right:** Ablation on growth factor α and initial decay d_0 . With a small d_0 , a positive growth factor ($\alpha > 0$) is required to adapt to increasing response lengths and maintain a smooth intra-sequence schedule. While a large d_0 ensures smoothness even with $\alpha = 0$, it limits exploration, resulting in lower learning efficiency.

Reverse Annealing We also test a reverse schedule, dubbed “RevEAD”, where temperature increases during generation: $\tau_t = \min\{\tau_{\min} + e^{t/d}, \tau_{\max}\}$. As shown in Fig. 10, RevEAD fails to even outperform standard temperature sampling, validating the necessity of an annealing (decreasing) temperature schedule. This observation also supports our hypothesis that late-stage exploration can be detrimental, potentially leading to high-variance “lucky” successes rather than robust reasoning paths.

Hyperparameter Sensitivity. We conduct an ablation study on the decay cap $d_{\max}$ and the growth factor α . As shown in Fig. 11a, EAD is robust to a wide range of $d_{\max}$ values, provided the cap is sufficiently large to ensure a smooth annealing schedule. Conversely, a very small cap (e.g., $d_0 = d_{\max} = 25$) approximates a fixed schedule (equivalent to $\alpha = 0$ throughout training). This setting causes an abrupt performance drop, likely because the model cannot adapt to the increasing response lengths during training (see § D). Alter-

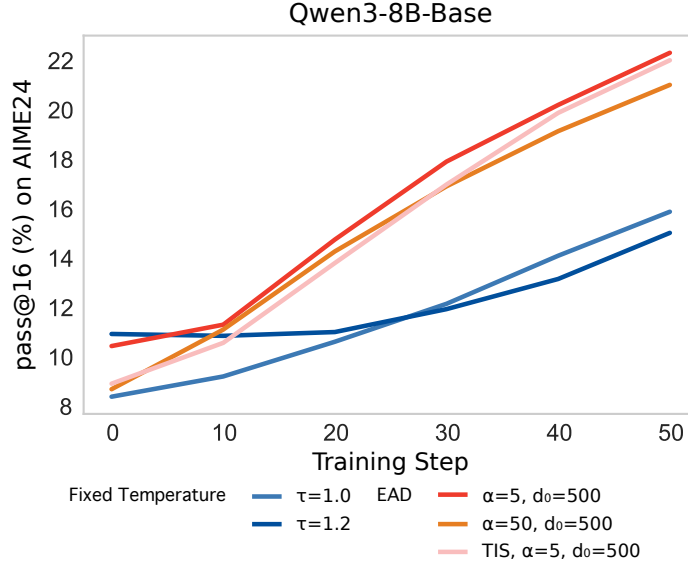


Figure 12: Stress-test of EAD using the DAPO recipe on AIME24. EAD outperforms the temperature sampling baseline across various hyperparameter setups.

natively, using a large initial d_0 yields a smooth schedule even with $\alpha = 0$. However, while training remains stable, performance improves slowly due to overly conservative exploration, as shown in Fig. 11b.

Intra-Sequence vs. Training-Step Scheduling. A natural question is whether the gains come from the *intra-sequence* annealing or merely from changing the effective temperature across training steps (via the global-step-aware decay rate). The ablation with $d_0 = d_{\max} = 25$ (Fig. 11a) isolates this: it retains the intra-sequence schedule but removes the training-step adaptation, and still outperforms the best fixed-temperature baseline, confirming that the within-sequence temperature schedule is the primary driver. The global-step-aware component provides further gains by keeping the schedule well-calibrated as response lengths grow.

5.6 Stress Test: DAPO-17K Recipe for AIME24

To verify the effectiveness of EAD on realistic mathematical reasoning tasks, we stress-test it using the DAPO recipe (Yu et al., 2025), training on DAPO-17k and evaluating on the challenging AIME 2024 benchmark. To accommodate GPU memory constraints, we utilize the Qwen-3-8B-Base (Yang et al., 2025) model (instead of 32B) and reduce the maximum response length from 20480 to 8192. We set $d_0 = 500$ and $d_{\max} = 40000$, and vary $\alpha \in \{5, 50\}$. As shown in Fig. 12, EAD achieves improved learning efficiency and consistently outperforms the standard temperature sampling baseline. We note that this experiment was conducted under reduced compute (8B model, shorter max length) and was not trained to full convergence; the results primarily demonstrate that EAD accelerates learning, with the final performance ceiling under full-scale training left for future work.

6 Related Work

Reinforcement Learning with Verifiable Rewards. Recent large-scale reasoning models such as OpenAI o1 (OpenAI, 2024), DeepSeek-R1 (Guo et al., 2025) have demonstrated that reinforcement-learning-based post-training can substantially enhance LLM reasoning. Motivated by this reinforcement learning with verifiable rewards (RLVR) (e.g. Shao et al. (2024); Guo et al. (2025); Lambert et al. (2024); Yang et al. (2025); Hu et al. (2025); Yu et al. (2025); Guan et al. (2025); Zeng et al. (2025) among many other works) has become a major approach for post-training LLMs to improve reasoning. A broad literature studies how to make RLVR training effective and efficient at scale, including novel reinforcement learning

algorithms and objectives (Yu et al., 2025; Liu et al., 2025b; Yue et al., 2025b; Zheng et al., 2025a), verifier architecture and reward designs (Zuo et al., 2025; Zhao et al., 2025b; Agarwal et al., 2025; Prabhudesai et al., 2025), and mechanisms that manage exploration diversity and entropy (Cheng et al., 2025; Chen et al., 2025; Cui et al., 2025; Wang et al., 2025b), or takes a critical view on the current evaluation (Yue et al., 2025a; Zhao et al., 2025a; Hochlehnert et al., 2025). Despite steady progress, a fundamental challenge is to balance exploration and exploitation along long reasoning trajectories without brittle heuristics. We adopt a simple yet effective annealed sampling schedule that front-loads exploration and cools later steps during rollout to encourage exploration while keeping training stable.

Exploration Control in RLVR. A line of works that is close to our work studies how to control exploration and sampling while doing RLVR (Hou et al., 2025; Cheng et al., 2025; Tang et al., 2025; Chen et al., 2025; Cui et al., 2025; Wang et al., 2025b; Xiong et al., 2025; Deng et al., 2025; Xu et al., 2025; Zheng et al., 2025b; Dou et al., 2025; Li et al., 2025). In particular, Cheng et al. (2025) propose per-token entropy to focus exploration at branching tokens in sampling; Tang et al. (2025); Chen et al. (2025) transform per-prompt rewards to optimize pass@k, guiding exploration across samples; Cui et al. (2025) control high-covariance tokens to prevent entropy collapse and sustain exploration; Wang et al. (2025b) update only high-entropy tokens, concentrating exploration where decisions split. While more nuanced multi-threaded exploration strategies (Pan et al., 2025) could benefit training, adapting them to the training loop for large models often introduces significant computational overhead and the challenges of maintaining massive parallelization. Different from prior work, this paper presents the first systematic analysis of sampling temperature and introduce a purely sampling-level annealed schedule that encourages exploration and then progressively stabilize answers, thereby enabling discovery of new solutions while yielding more stable training.

Simulated Annealing. Simulated Annealing (SA) is a probabilistic optimization technique inspired by annealing in metallurgy, designed to find the global optimum in a large search space (Kirkpatrick et al., 1983; Bertsimas & Tsitsiklis, 1993). The core principle involves a temperature parameter that controls the probability of accepting suboptimal states. Initially, a high temperature allows the search to escape local minima by exploring broadly (exploration). As the temperature gradually decreases, the algorithm increasingly favors better states, converging towards a high-quality solution (exploitation). This “coarse-to-fine” search dynamic, where high temperatures establish a solution’s general structure and low temperatures refine its details, strongly parallels the generative process of LLMs (Yang & Holtzman, 2025). SA has been adapted in various machine learning contexts to manage the exploration-exploitation trade-off, including recent applications in graph optimization (Liu et al., 2021), text editing (Zhang et al., 2024), non-autoregressive generation (Israel et al., 2025), and efficient Best-of-N sampling (Manvi et al., 2024). However, these prior applications invariably apply a single, uniform temperature across all positions in a generated sequence. This approach fails to account for the heterogeneous roles of tokens at different positions (Wang et al., 2025b). Our work departs from this convention. To the best of our knowledge, we are the first to introduce an **intra-sequence annealed temperature** schedule, where the temperature varies dynamically within the generation of a single sequence. This novel approach allows for more nuanced control over exploration and leads to significant performance gains in RLVR.

7 Discussion

Our work addresses a central challenge in RLVR: achieve an effective balance between exploration and exploitation. We introduce Exploratory Annealed Decoding (EAD), a simple yet powerful sampling strategy that avoids heavy computation and intricate heuristics. Specifically, EAD employs a temperature-annealing schedule that begins with a high sampling temperature and gradually cools, enabling LLMs to explore broadly at the beginning of generation and converge toward precise, high-quality completions throughout the decoding process. As RLVR often relies on multiple rollouts to estimate rewards, this annealing schedule effectively improves sampling diversity while controlling variance, making it well suited for RL training. At the same time, EAD can also be applied directly at test time to

enhance inference efficiency and scaling, improving the quality of single- or multi-sample decoding without additional computation cost.

While EAD is conceptually simple, this simplicity is a deliberate design choice to ensure scalability and ease of adoption. Unlike complex strategies that require multi-threading (Pan et al., 2025) or tracking additional statistics (Cui et al., 2025), EAD adds zero computational overhead and integrates seamlessly with standard training pipelines. Our results demonstrate that this lightweight approach consistently outperforms baselines across diverse models and RL algorithms, suggesting that effective exploration need not come at the cost of system complexity.

Despite the encouraging results, our study has several limitations that suggest directions for future work. First, our evaluation focuses on mathematical reasoning with verifiable rewards across diverse model families (LLaMA and Qwen) and sizes (1B–8B). While the “explore early, exploit late” principle is grounded in a general property of autoregressive generation (§ 3), evaluating EAD on other verifiable domains (e.g., code generation) and at larger scales remains an important direction. Second, our primary evaluation uses multi-sample metrics (Pass@ k , Worst@ k , Majority@ N), which are the natural metrics for RLVR since it inherently relies on generating multiple rollouts per prompt. Single-sample metrics (e.g., Pass@1) are less informative in this context, as the variance of a single rollout provides a noisy signal for both training and evaluation. Investigating the interplay between exploration strategies and single-path generation quality, as well as combining EAD with other advanced exploration-promoting algorithms (see § 6), are promising directions. Third, our experiments adopt a uniform temperature schedule across all prompts. Although an adaptive, prompt-specific schedule could further enhance performance, developing such a mechanism is nontrivial—collecting extra statistics (e.g., token-wise entropy quantile (Wang et al., 2025b) or probability-advantage covariance (Cui et al., 2025)) adds computational overhead and system complexity (Li et al., 2025; Liu et al., 2025a). We leave adaptive scheduling for future investigation.

In summary, EAD provides a simple yet general way to couple exploration with the inherent progression of language generation. By reducing algorithmic overhead while improving trajectory quality, it opens new avenues for both efficient inference and effective reinforcement fine-tuning.

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633 A Minimal-RL Training Details

634 We mainly follow the Minimal-RL recipe (Xiong et al., 2025) in our experiments to ensure a
 635 fair comparison among different rollout sampling strategies. Specifically, we set a series of
 636 hyperparameters as in Table 1:

Hyperparameter	Value(s)
Training Batch Size	1024
Max Prompt Length	1024
Max Response Length	3072
Mini Batch Size	256
Micro Batch Size Per GPU	4
Learning Rate	10^{-6}

Table 1: Hyperparameter Setup for Running Minimal-RL recipe.

637 B Off-policy Issue and Truncated Importance Sampling Correction

638 B.1 Sampling Techniques Can Introduce Off-Policy Issue

639 One subtle yet troublesome drawback of reinforcement learning with sampling techniques is
 640 that it simultaneously introduces the *off-policy* problem: there is a gap between the behavior
 641 policy (used for sampling) and the target policy (being optimized and parametrized by θ).
 642 This might introduce instability to the training and cause it to fail (See example training of
 643 RLVR with EAD Fig. 13).

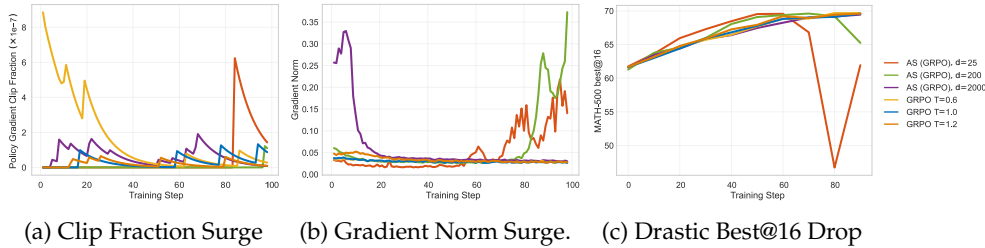


Figure 13: Off-policy samples bring training instability. The base model is Qwen2.5-Math-1.5B.

644 Noted that this off-policy phenomenon widely exists for any efficient sampling framework
 645 (Yao et al., 2025) and sampling strategy (for instance, when applying best-of- n sampling
 646 (Xiong et al., 2025) or filtering out responses (Shrivastava et al., 2025), the underlying
 647 distribution of responses is implicitly altered). To be more precise, we take the policy
 648 gradient loss as an example:

$$\mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta_{\text{old}}}^{\text{sampling}}(\cdot | x)} \left[\frac{\pi_{\theta}(y | x)}{\pi_{\theta_{\text{old}}}(y | x)} A(y; x) \right] = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta_{\text{old}}}(\cdot | x)} \left[\frac{\pi_{\theta_{\text{old}}}^{\text{sampling}}(y | x)}{\pi_{\theta_{\text{old}}}(y | x)} \times \frac{\pi_{\theta}(y | x)}{\pi_{\theta_{\text{old}}}(y | x)} A(y; x) \right],$$

where $\pi_{\theta_{\text{old}}}^{\text{sampling}}(\cdot | x)$ represents the underlying sampling distribution. In such case, an extra weight is implicitly added to each response in addition to its advantage $A(y; x)$.

This extra weight can significantly inflate the variance of the policy gradient, posing a stability challenge that our proposed EAD needs to mitigate. To quantify this effect in our proposed EAD, we now analyze such variance under a standard, fixed **temperature sampling**.

We use $\tau = 1$ to define a policy π and consider the effect of τ on the variance of the gradient estimator. We reduce the problem to analyzing the variance inflation factor

$$\mathbb{E}_{y \sim \pi(\cdot | x; \tau)} \left[\frac{\pi(y | x; 1)^2}{\pi(y | x; \tau)^2} \right]. \quad (2)$$

We begin with one-token case. Let o_i denote the i th token in the vocabulary V and h_i is its logit. Then equation 2 can be rewritten as

$$\sum_{i=1}^{|V|} \frac{h_i / (\sum_{j=1}^{|V|} h_j)}{h_i^{1/\tau} / (\sum_{j=1}^{|V|} h_j^{1/\tau})} \times \frac{h_i}{\sum_{j=1}^{|V|} h_j} = \frac{\sum_{i=1}^{|V|} h_i^{2-1/\tau} \sum_{i=1}^{|V|} h_i^{1/\tau}}{\left(\sum_{i=1}^{|V|} h_i \right)^2}$$

Proposition B.1. Suppose $h_i \in [0, 1]$ for all $i \in V$. $\sum_{i=1}^{|V|} h_i^{2-1/\tau} \sum_{i=1}^{|V|} h_i^{1/\tau}$ is decreasing when $\tau \leq 1$ and increasing when $\tau \geq 1$, which implies it has a global minimum at $\tau = 1$.

Proof. Let $x = 1/\tau$. We define

$$f(x) = \log \left(\sum_{i=1}^{|V|} h_i^{2-x} \right) + \log \left(\sum_{i=1}^{|V|} h_i^x \right).$$

Its derivative is

$$f'(x) = \frac{-\sum_{i=1}^{|V|} h_i^{2-x} \log h_i}{\sum_{i=1}^{|V|} h_i^{2-x}} + \frac{\sum_{i=1}^{|V|} h_i^x \log h_i}{\sum_{i=1}^{|V|} h_i^x}.$$

To analyze the sign of $f'(x)$, we define a helper function $g(x) = \frac{\sum_{i=1}^{|V|} h_i^x \log h_i}{\sum_{i=1}^{|V|} h_i^x}$. Then, $f'(x) = g(x) - g(2-x)$ and its sign depends on whether $g(x)$ is greater than, less than, or equal to $g(2-x)$. We take a look at derivative of g :

$$g'(x) = \frac{\left(\sum_{i=1}^{|V|} h_i^x (\log h_i)^2 \right) \left(\sum_{i=1}^{|V|} h_i^x \right) - \left(\sum_{i=1}^{|V|} h_i^x \log h_i \right)^2}{\left(\sum_{i=1}^{|V|} h_i^x \right)^2} \geq 0.$$

Hence, g is an increasing function and

$$\begin{cases} f'(x) = g(x) - g(2-x) \geq 0, & \text{when } x \geq 1 \\ f'(x) = g(x) - g(2-x) = 0, & \text{when } x = 1 \\ f'(x) = g(x) - g(2-x) \leq 0, & \text{when } x \leq 1. \end{cases}$$

Accordingly, f is increasing when $x \geq 1$ and is decreasing when $x \leq 1$. Then the proposition easily follows. $\square$

The same conclusion can be proved for multiple-token sequence by induction. Therefore, we get that when the temperature is far from 1, the off-policy issue could be severe and lead to large variance of the gradient estimator.

672 B.2 Truncated Importance Sampling Ratio Correction

673 To cancel such bias, an importance sampling ratio can be introduced (Hilton et al., 2022; Yao
674 et al., 2025):

$$\mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta_{\text{old}}}(\cdot | x)} \left[\frac{\pi_{\theta}(y | x)}{\pi_{\theta_{\text{old}}}(y | x)} A(y; x) \right] = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta_{\text{old}}}^{\text{sampling}}(\cdot | x)} \left[\frac{\pi_{\theta_{\text{old}}}(y | x)}{\pi_{\theta_{\text{old}}}^{\text{sampling}}(y | x)} \times \frac{\pi_{\theta}(y | x)}{\pi_{\theta_{\text{old}}}(y | x)} A(y; x) \right].$$

675 To further prevent negative effects by the extreme likelihood ratios and boost training
676 stability, we truncate the likelihood ratio with an upper bound. That is, *truncated importance*
677 *sampling* technique (Heckman et al., 1998). Taking the vanilla policy gradient loss as an
678 example, the modified loss for EAD is as follows:

$$\mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta_{\text{old}}}^{\text{EAD}}(\cdot | x)} \left[\min \left(\frac{\pi_{\theta_{\text{old}}}(y | x)}{\pi_{\theta_{\text{old}}}^{\text{EAD}}(y | x)}, \varepsilon \right) \frac{\pi_{\theta}(y | x)}{\pi_{\theta_{\text{old}}}(y | x)} A(y; x) \right].$$

C Proof of Temperature-Entropy Relationship

Proposition C.1. *The entropy of the softmax distribution is a non-decreasing function of the temperature $\tau > 0$.*

Proof. The strategy is to show that the entropy H is a non-increasing function of the inverse temperature $\beta = 1/\tau > 0$. The probability of sampling token v with temperature τ is given by the policy π_θ . For simplicity in the derivation, we denote this probability as $p_v(\tau)$:

$$p_v(\tau) \triangleq \pi_\theta(y_t = v \mid [x, y_{<t}]; \tau)$$

Let h_v be the logit for a token v in the vocabulary V . The probability of a token as a function of β is given by:

$$p_v(\beta) = \frac{\exp(\beta h_v)}{\sum_{v' \in V} \exp(\beta h_{v'})} \triangleq \frac{\exp(\beta h_v)}{Z(\beta)},$$

where $Z(\beta)$ is the partition function. The entropy, as a function of β , is:

$$H(\beta) = - \sum_{v \in V} p_v(\beta) \log p_v(\beta).$$

We can rewrite the entropy by substituting $\log p_v(\beta) = \beta h_v - \log Z(\beta)$:

$$\begin{aligned} H(\beta) &= - \sum_{v \in V} p_v(\beta) (\beta h_v - \log Z(\beta)) \\ &= \log Z(\beta) \left(\sum_{v \in V} p_v(\beta) \right) - \beta \sum_{v \in V} h_v p_v(\beta) \\ &= \log Z(\beta) - \beta \cdot \mathbb{E}_{v \sim p(\beta)}[h_v]. \end{aligned}$$

Now, we differentiate $H(\beta)$ with respect to β . Let $\bar{h}(\beta) = \mathbb{E}[h_v]$.

$$\frac{dH}{d\beta} = \frac{d}{d\beta}(\log Z(\beta)) - \frac{d}{d\beta}(\beta \bar{h}(\beta)).$$

First, we find the derivative of the log-partition function:

$$\frac{d}{d\beta}(\log Z(\beta)) = \frac{Z'(\beta)}{Z(\beta)} = \frac{\sum_v h_v \exp(\beta h_v)}{Z(\beta)} = \sum_v h_v p_v(\beta) = \bar{h}(\beta).$$

Next, we use the product rule for the second term:

$$\frac{d}{d\beta}(\beta \bar{h}(\beta)) = \bar{h}(\beta) + \beta \frac{d\bar{h}}{d\beta}.$$

Combining these gives:

$$\frac{dH}{d\beta} = \bar{h}(\beta) - \left(\bar{h}(\beta) + \beta \frac{d\bar{h}}{d\beta} \right) = -\beta \frac{d\bar{h}}{d\beta}.$$

The derivative $\frac{d\bar{h}}{d\beta}$ is the variance of the logits. We can show this by differentiating $\bar{h}(\beta)$:

$$\begin{aligned} \frac{d\bar{h}}{d\beta} &= \frac{d}{d\beta} \left(\frac{\sum_v h_v \exp(\beta h_v)}{Z(\beta)} \right) \\ &= \frac{(\sum_v h_v^2 \exp(\beta h_v))Z(\beta) - (\sum_v h_v \exp(\beta h_v))Z'(\beta)}{Z(\beta)^2} \\ &= \sum_v h_v^2 p_v(\beta) - \left(\sum_v h_v p_v(\beta) \right) \left(\frac{Z'(\beta)}{Z(\beta)} \right) \\ &= \mathbb{E}[h^2] - (\mathbb{E}[h])^2 = \text{Var}_{v \sim p(\beta)}(h_v). \end{aligned}$$

694 Substituting this back, we arrive at the final expression for the derivative of entropy:

$$\frac{dH}{d\beta} = -\beta \cdot \text{Var}_{v \sim p(\beta)}(h_v).$$

695 By definition, the temperature $\tau > 0$, so the inverse temperature $\beta > 0$. The variance of any
 696 random variable is non-negative. This can be formally shown using **Jensen’s inequality**: for
 697 the convex function $\phi(x) = x^2$, we have $\mathbb{E}[\phi(h)] \geq \phi(\mathbb{E}[h])$, which means $\mathbb{E}[h^2] \geq (\mathbb{E}[h])^2$,
 698 and thus $\text{Var}(h) \geq 0$.

699 Therefore, the derivative of entropy with respect to β is non-positive:

$$\frac{dH}{d\beta} = \underbrace{-\beta}_{\leq 0} \cdot \underbrace{\text{Var}(h_v)}_{\geq 0} \leq 0.$$

700 Since $H(\beta)$ is a non-increasing function of β , and β is inversely proportional to T , it follows
 701 that $H(\tau)$ must be a non-decreasing function of the temperature τ . $\square$

702 D Increasing Length in RL Training

703 During RL training, our algorithm (EAD) incentivizes the model to generate longer, more
 704 effective reasoning chains for difficult problems, especially for 7B models (Fig. 14). While
 705 both EAD and temperature sampling initially learn to shorten their responses by shifting
 706 from complex code-based solutions to direct mathematical reasoning, their behavior later
 707 diverges. The response length from temperature sampling stabilizes, whereas EAD learns to
 708 selectively increase reasoning length for harder problems, which boosts final performance.
 709 For these experiments, EAD is applied in the DAPO algorithm for sampling rollouts. We
 710 use the same training setup as detailed in § 5.

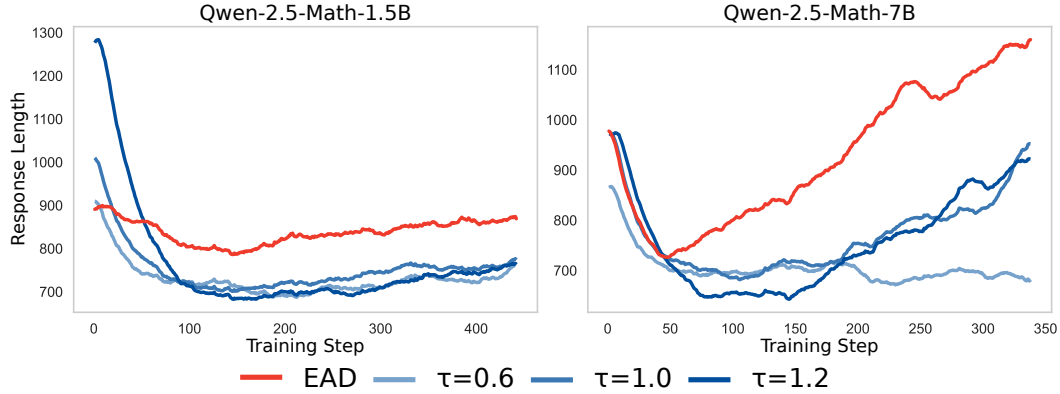


Figure 14: Compared with normal temperature sampling, EAD can naturally incentivize the model to generate longer reasoning chains.